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EN3160 Tutorial 2

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1. Total least squares is useful in vision problems.

- (a) Show how to find the line parameters expressed as $ax + by = d$ using this method in terms of point coordinates (x_i, y_i) , $i = 1, \dots, N$, and U , in regular notation.

[illegible]

- (b) In a particular case, $\bar{x} = 5.0$, $\bar{y} = 13.1$. The eigenvalues of $U^T U$ are 21.2, 4324.1 and corresponding eigenvectors are $[-0.8974, 0.4411]^T$, and $[-0.4411, -0.8974]^T$. Find the gradient m and intercept c of the total-least-squares-fit line.

A series of horizontal dotted lines for writing.

2. Figure 1 shows the image 1 and 4 from the graffiti images, and the homography that maps image 1 to 4. Note the convention of axes.

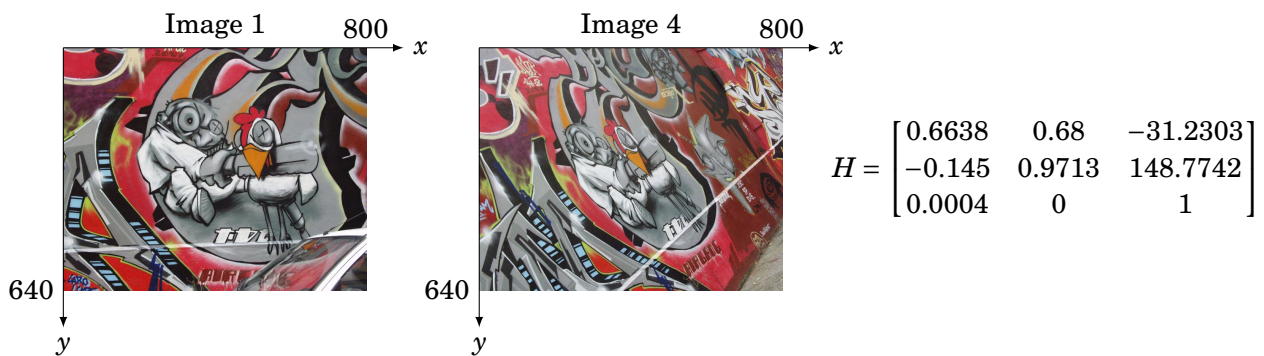


Figure 1: Image 1, image 4 and homography mapping image 1 to 4.

- (a) State the transformation that maps image 4 to image 1.

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- (b) Compute the locations of the corners of image 4 when mapped onto image 1.

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- (c) What is the size of the arrays needed to accommodate the stitched image after this mapping?

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- (d) What are two additional steps to make the stitched result visually pleasing?

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3. The scale-normalized Laplacian of Gaussian (LoG) for circularly symmetric blob detection is

$$\nabla^2_{\text{norm}}g = \sigma^2 \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \right),$$

where g is the bi-variate Gaussian.

- (a) Obtain an expression for the radius of a black circle in a white background that would give the maximum response to this operator.
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- (b) What is the suitable value for σ , if black circles of radius 10 pixels are to be detected.
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- (c) Assume that two images of the scene—taken with two level of zoom—have features detected. How can the features be matched?
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4. A conveyor belt carries *several types* wrapped chocolate bars. Images of reference wrapped chocolate bars are available. The factory has the following requirements:

- Counting the number of wrapped chocolate bars that pass along the belt.
- Determining if there are printing mistakes in comparison with references.

- (a) Propose a system for counting.
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- (b) Propose a method for locating printing mistakes.
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5. Consider the set of points that corresponds to a part of a corner of a box

$$X = \begin{bmatrix} 0 & 1.277 \\ 0 & -0.796 \\ 3 & -2.016 \\ 3 & 2.196 \\ 6 & -2.907 \\ 6 & 6.594 \\ 9 & -7.897 \\ 9 & 6.373 \\ 12 & -10.023 \\ 12 & 12.723 \end{bmatrix}$$

also shown in shown in Fig. 2. The figure includes a least-squares fitted line and thresholds used to determine the consensus.

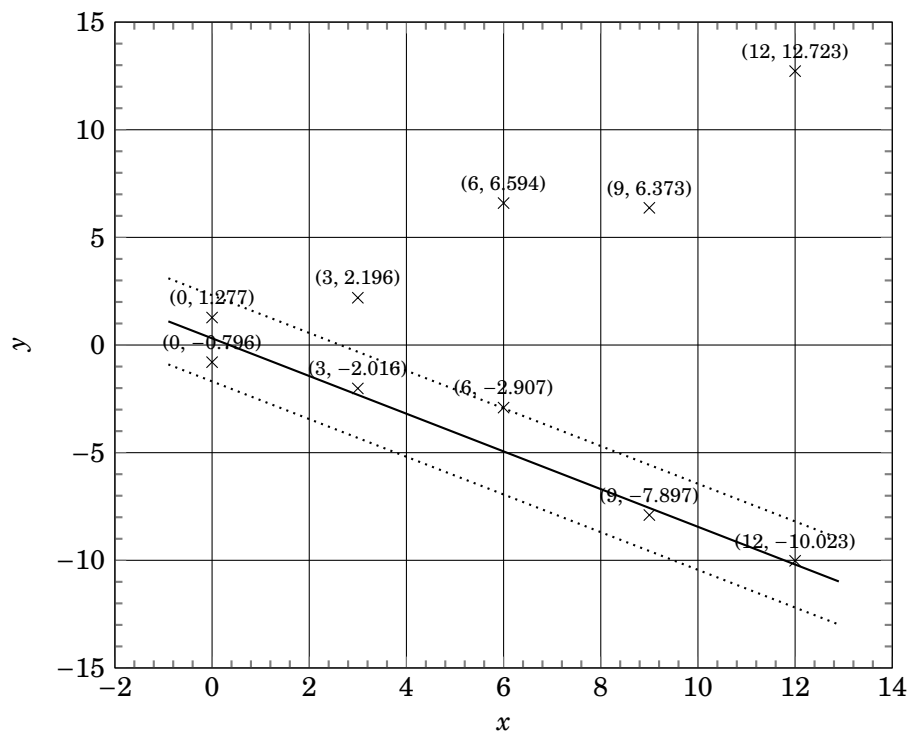


Figure 2: Points for line fitting.

(a) State how a second line can be fitted.

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(b) Using the RANSAC method, find an expression for this line. You may hand-pick the sample by observation. Run only one iteration. Use sketches to illustrate.

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(c) Briefly describe the process of fitting a circle and lines to a (different) set of points.

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(d) The third edge of the box is vertical and has generated a set of noisy points. How would you fit a line to these points?

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6. Consider the scale space shown in Fig. 3.

(a) Identify the scale-space extrema by observation, and mark them in the figure.

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(b) Compute the radii of the blobs that correspond to each of these extrema.

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(c) What must be done to be able to match images of the same object taken from different viewpoints?

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7. In computing the homography using point matches between two images through RANSAC, there were 1600 inliers.

(a) What is the minimum number of point matches required to compute the homography?

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(b) Show the format of the “data matrix” used for homography computation using singular value decomposition (SVD).

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-8	-18	-45	-75	-112	-101	-53	-101	-112	-75	-45	-18	-5	-2	-2	-2	-2	-2	-2	-2
-18	-34	-49	-44	-63	-44	9	-44	-63	-44	-49	-33	-10	-3	-2	-2	-2	-2	-2	-2
-45	-49	-11	55	56	51	71	51	56	55	-11	-46	-23	-5	-2	-2	-2	-2	-2	-2
-75	-44	55	93	64	42	39	42	64	93	55	-37	-39	-9	-2	-2	-2	-2	-2	-2
-112	-63	56	64	21	8	6	8	21	64	57	-50	-57	-15	-3	-2	-2	-2	-2	-2
-101	-44	51	42	8	0	0	0	8	42	53	-25	-52	-21	-4	-2	-2	-2	-2	-2
-53	9	71	39	6	0	0	0	6	39	75	29	-28	-22	-5	-2	-2	-2	-2	-2
-101	-44	51	42	8	0	0	0	8	42	53	-25	-52	-21	-4	-2	-2	-2	-2	-2
-112	-63	56	64	21	8	6	8	21	64	57	-50	-57	-15	-3	-4	-5	-4	-3	-2
-75	-44	55	93	64	42	39	42	64	93	55	-37	-39	-12	-13	-20	-22	-20	-13	-6
-45	-49	-11	55	57	53	75	53	57	55	-11	-47	-29	-25	-44	-46	-26	-46	-44	-23
-18	-33	-46	-37	-50	-25	29	-25	-50	-37	-47	-36	-30	-39	-35	-10	35	-10	-35	-42
-5	-10	-23	-39	-57	-52	-28	-52	-57	-39	-24	-21	-46	-35	50	78	85	78	49	-46
-2	-3	-5	-9	-15	-21	-22	-21	-15	-9	-7	-21	-47	-10	78	75	51	75	76	-28
-2	-2	-2	-2	-3	-4	-5	-4	-3	-2	-5	-22	-26	35	85	51	23	51	82	15
-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-4	-20	-46	-10	78	75	51	75	76	-28
-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-3	-13	-44	-35	50	78	85	78	49	-46
-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-6	-22	-39	-35	-10	35	-10	-35	-42	-43
-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-3	-8	-22	-44	-46	-26	-46	-44	-23	-14
-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-3	-6	-13	-20	-22	-20	-13	-6	-4
-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	-3	-6	-8	-6	-3	-2

(a) $\sigma = 1$.

-137	-129	-105	-75	-45	-24	-13	-14	-23	-39	-56	-69	-73	-68	-57	-44	-30	-20	-14	-10
-129	-120	-95	-64	-34	-12	-1	-1	-12	-29	-47	-62	-68	-66	-56	-43	-30	-20	-14	-10
-105	-95	-69	-35	-2	21	32	31	20	0	-23	-44	-57	-60	-54	-43	-31	-21	-15	-11
-75	-64	-35	3	37	62	73	73	60	37	7	-21	-43	-53	-53	-45	-34	-24	-17	-13
-45	-34	-2	37	73	99	111	110	97	70	35	-1	-30	-48	-53	-48	-39	-29	-22	-18
-24	-12	21	62	99	125	137	136	122	93	54	12	-23	-47	-56	-53	-45	-37	-30	-25
-13	-1	32	74	111	137	150	149	133	102	60	16	-23	-49	-60	-60	-53	-45	-39	-35
-14	-1	31	73	111	137	149	147	130	98	55	10	-29	-55	-65	-64	-58	-52	-48	-46
-23	-12	20	60	97	122	134	130	112	80	39	-4	-38	-58	-65	-62	-56	-52	-51	-52
-39	-29	0	37	71	94	104	100	81	51	14	-21	-45	-57	-55	-48	-42	-41	-45	-50
-56	-47	-23	8	37	56	64	59	42	16	-13	-36	-48	-45	-33	-20	-12	-15	-25	-35
-69	-62	-44	-21	1	16	21	15	1	-19	-37	-47	-42	-25	0	21	30	23	7	-9
-73	-68	-57	-41	-27	-18	-16	-21	-32	-45	-53	-50	-31	1	37	64	75	65	43	21
-68	-66	-60	-51	-44	-40	-40	-45	-53	-60	-60	-46	-17	24	67	99	110	97	71	45
-57	-56	-54	-51	-48	-47	-49	-54	-60	-64	-59	-41	-7	37	82	114	125	110	82	55
-44	-43	-43	-43	-42	-44	-47	-51	-56	-59	-54	-37	-6	35	76	104	112	97	69	43
-30	-30	-31	-32	-33	-35	-38	-42	-47	-51	-48	-36	-13	19	51	73	78	63	38	15
-20	-20	-21	-22	-23	-25	-28	-32	-38	-43	-44	-39	-25	-6	16	30	31	18	-3	-22
-14	-14	-14	-15	-16	-17	-20	-24	-30	-37	-42	-43	-39	-30	-19	-13	-15	-26	-43	-57
-10	-10	-10	-11	-11	-12	-15	-19	-25	-33	-40	-46	-49	-47	-44	-44	-48	-58	-71	-82
-9	-9	-9	-9	-9	-10	-12	-16	-23	-31	-40	-47	-52	-54	-53	-54	-59	-69	-81	-91

(c) $\sigma = 3$.

-115	-110	-99	-86	-75	-66	-62	-65	-74	-80	-82	-76	-61	-40	-23	-12	-8	-6	-6	-6
-110	-101	-81	-61	-46	-36	-33	-36	-44	-54	-63	-66	-58	-41	-24	-13	-8	-6	-6	-6
-99	-81	-42	-4	18	27	28	27	21	5	-20	-44	-53	-44	-28	-15	-9	-6	-6	-6
-86	-61	-4	47	71	75	73	75	74	59	23	-20	-46	-47	-33	-18	-10	-7	-6	-6
-75	-46	18	71	89	86	81	86	94	86	49	-3	-41	-49	-37	-21	-11	-7	-6	-6
-66	-36	27	75	86	76	68	76	91	91	59	7	-36	-49	-40	-24	-13	-8	-7	-7
-62	-33	28	73	81	68	60	69	87	90	61	9	-36	-52	-43	-28	-17	-12	-11	-10
-65	-36	27	75	86	76	69	77	92	91	58	4	-41	-58	-51	-37	-26	-21	-19	-18
-74	-44	21	74	94	91	87	92	98	88	48	-9	-53	-68	-60	-47	-38	-34	-33	-32
-80	-54	5	59	86	91	90	91	88	68	24	-29	-66	-75	-64	-49	-40	-40	-45	-50
-82	-63	-20	23	49	59	62	59	50	26	-13	-53	-75	-71	-50	-27	-18	-24	-43	-61
-76	-66	-44	-20	-3	7	10	7	-4	-23	-49	-70	-73	-49	-11	23	35	20	-16	-53
-61	-58	-53	-46	-41	-36	-34	-36	-44	-55	-68	-73	-57	-14	42	87	102	79	27	-29
-40	-41	-44	-47	-49	-49	-49	-50	-54	-59	-65	-62	-35	20	87	138	154	127	65	-1
-23	-24	-28	-33	-37	-39	-39	-40	-42	-47	-53	-49	-22	35	105	158	173	145	80	-21
-12	-13	-15	-18	-21	-23	-23	-24	-26	-32	-41	-42	-22	28	90	139	154	127	65	-1
-8	-8	-9	-10	-11	-12	-12	-12	-15	-21	-32	-39	-30	3	50	90	102	79	27	-29
-6	-6	-6	-7	-7	-7	-7	-7	-10	-15	-25	-36	-38	24	2	28	36	20	-17	-54
-6	-6	-6	-6	-6	-6	-6	-6	-8	-12	-20	-30	-39	-41	-34	-24	-20	-29	-47	-64
-6	-6	-6	-6	-6	-6	-6	-6	-7	-9	-15	-24	-35	-44	-49	-50	-51	-53	-58	-62
-6	-6	-6	-6	-6	-6	-6	-6	-7	-9	-13	-21	-32	-44	-53	-58	-59	-60	-60	-59

(b) $\sigma = 2$.

-97	-91	-73	-48	-21	2	16	19	12	-2	-21	-40	-55	-65	-67	-63	-55	-45	-37	-32
-91	-84	-66	-41	-14	9	23	26	19	3	-16	-36	-53	-63	-66	-63	-55	-46	-38	-33
-73	-66	-48	-22	5	28	42	44	36	18	-4	-26	-46	-59	-64	-63	-57	-49	-42	-37
-48	-41	-22	5	33	55	69	70	60	40	14	-12	-36	-53	-62	-63	-59	-53	-47	-42
-21	-14	5	33	61	84	96	97	84	61	32	2	-26	-47	-60	-64	-62	-57	-52	-48
2	9	28	56	84	107	119	118	103	78	46	13	-18	-42	-57	-64	-64	-61	-57	-54
16	23	42	69	97	119	130	128	113	86	53	19	-13	-38	-55	-62	-64	-62	-59	-57
20	26	45	71	98	119	129	126	111	85	52	19	-12	-35	-50	-58	-59	-58	-56	-54
13	19	37	61	86	105	114	111	96	73	43	14	-12	-31	-43	-48	-49	-48	-47	-46
-1	4	20	42	64	81	88	86	73	53	29	6	-13	-25	-32	-33	-32	-31	-30	-31
-20	-15	-2	17	36	50	56	54	44	29	12	-2	-12	-17	-16	-12	-9	-7	-7	-8
-38	-34	-24	-9	6	17	22	21	14	5	-4	-10	-10	-6	2	11	17	20	19	18
-53	-50	-42	-31	-20	-12	-9	-10	-13	-17	-19	-16	-8	5	19	33	41	44	43	41
-62	-60	-55	-47	-40	-35	-33	-33	-35	-35	-31	-21	-6	13	32	48	58	61	60	57
-64	-63	-60	-55	-51	-49	-49	-49	-49	-46	-38	-25	-6	16	37	54	64	66	64	60
-60	-59	-58	-55	-54	-54	-55	-56	-56	-52	-43	-29	-10	12	32	48	57	58	55	51
-52	-51	-51	-50	-51	-52	-55	-57	-57	-54	-46	-34	-17	2	19	32	38	38	34	29
-42	-42	-42	-43	-45	-47	-51	-54	-55	-54	-49	-39	-26	-12	1	10	14	12	7	2
-34	-34	-35	-36	-39	-42	-46	-50	-53	-53	-51	-45	-36	-26	-17	-11	-11	-14	-20	-25
-28	-28	-29	-31	-34	-38	-42	-47	-51	-53	-52	-49	-43	-36	-30	-27	-28	-33	-39	-45
-26	-27	-28	-29	-32	-36	-41	-46	-50	-52	-53	-50	-46	-40	-35	-33	-35	-40	-46	-52

(d) $\sigma = 4$.

Figure 3: Scale space.

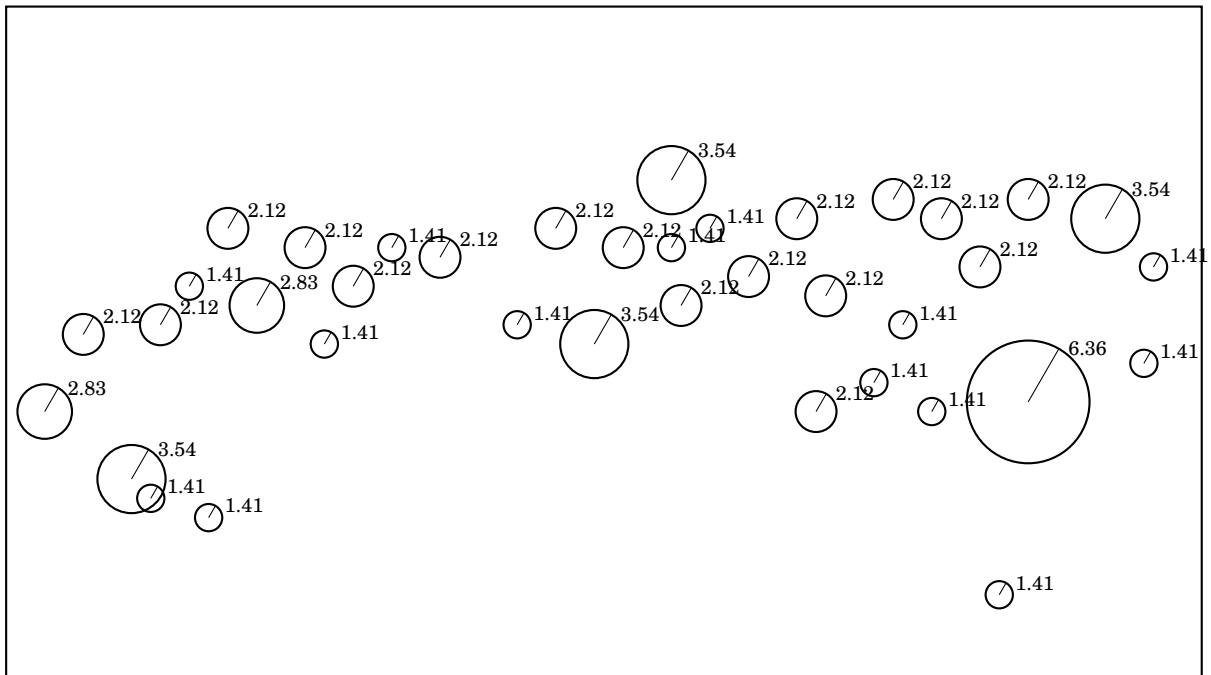
(c) When SVD is used the resulting $V^\top$ is

$$V^\top = \begin{bmatrix} 1.0 \times 10$$

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[illegible]

- ### Detected Extrema in Scale-Space



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(a) What is the reason for circles having a few specific radii?

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(b) Some circles fail to fit well to the central back disk of the flower. How can this be corrected?

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(c) In scale-space extrema detection, a large number of extrema are detected. Suggest a remedy for this.

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